

Integration by Substitution

ANSWER KEY



u-substitution

1 Evaluate each integral using substitution:

a $\int 2x(x^2 + 1)^5 dx$ $u = x^2 + 1$: $= \frac{(x^2 + 1)^6}{6} + C$.

b $\int \frac{x}{\sqrt{x^2 + 9}} dx$ $u = x^2 + 9$: $= \sqrt{x^2 + 9} + C$.

c $\int \cos(3x) dx$ $u = 3x$: $= \frac{1}{3} \sin(3x) + C$.

2 Evaluate $\int x^2 e^{x^3} dx$.

$u = x^3$, $du = 3x^2 dx$: $= \frac{1}{3} e^{x^3} + C$.

Definite integrals with substitution

3 Evaluate $\int_0^2 x \sqrt{x^2 + 1} dx$ exactly, changing the limits with $u = x^2 + 1$.

u goes from 1 to 5: $\frac{1}{2} \int_1^5 \sqrt{u} du = \frac{1}{2} \left[\frac{2}{3} u^{3/2} \right]_1^5 = \frac{1}{3} (5\sqrt{5} - 1)$.

4 Evaluate $\int_0^{\pi/2} \sin^3 x \cos x dx$.

$u = \sin x$, limits 0 to 1: $\int_0^1 u^3 du = \frac{1}{4}$.

5 Evaluate $\int_1^{e^{\ln x}} \frac{1}{x} dx$.

$u = \ln x$, limits 0 to 1: $\int_0^1 u du = \frac{1}{2}$.

Substitution with trigonometric and exponential integrands

6 Evaluate each integral using substitution:

a $\int \sin^4 x \cos x dx$ $u = \sin x$: $= \frac{\sin^5 x}{5} + C$.

b $\int \frac{e^x}{1 + e^x} dx$ $u = 1 + e^x$: $= \ln|1 + e^x| + C$.

c $\int \sec^2 x \tan x dx$ $u = \tan x$: $= \frac{\tan^2 x}{2} + C$.

7 Evaluate $\int \frac{2x + 3}{x^2 + 3x + 1} dx$, recognizing the numerator as the derivative of the denominator.

$u = x^2 + 3x + 1$, $du = (2x + 3) dx$: $= \ln|x^2 + 3x + 1| + C$.

Choosing the substitution

8 For each integral, state a suitable substitution u and evaluate:

a $\int x^2(x^3 - 1)^7 dx$ $u = x^3 - 1$, $du = 3x^2 dx$: $= \frac{1}{3} \int u^7 du = \frac{(x^3 - 1)^8}{24} + C$.

b $\int \tan x dx = \int \frac{\sin x}{\cos x} dx$ $u = \cos x$, $du = -\sin x dx$: $= -\ln|\cos x| + C = \ln|\sec x| + C$.

Substitution with radicals

9 Evaluate $\int x\sqrt{2-x} dx$ using the substitution $u = 2 - x$ (so $x = 2 - u$).

$u = 2 - x$, $x = 2 - u$, $dx = -du$:

$$\int (2-u)\sqrt{u}(-du) = \int (u-2)\sqrt{u} du = \int (u^{3/2} - 2u^{1/2}) du = \frac{2}{5}u^{5/2} - \frac{4}{3}u^{3/2} + C = \frac{2}{5}(2-x)^{5/2} - \frac{4}{3}(2-x)^{3/2} + C$$

10 Evaluate $\int_0^3 \frac{x}{\sqrt{x+1}} dx$ using $u = x + 1$ (so $x = u - 1$), changing the limits.

u goes from 1 to 4:

$$\int_1^4 \frac{u-1}{\sqrt{u}} du = \int_1^4 (u^{1/2} - u^{-1/2}) du = \left[\frac{2}{3}u^{3/2} - 2u^{1/2} \right]_1^4 = \left(\frac{16}{3} - 4 \right) - \left(\frac{2}{3} - 2 \right) = \frac{4}{3} + \frac{4}{3} = \frac{8}{3}.$$